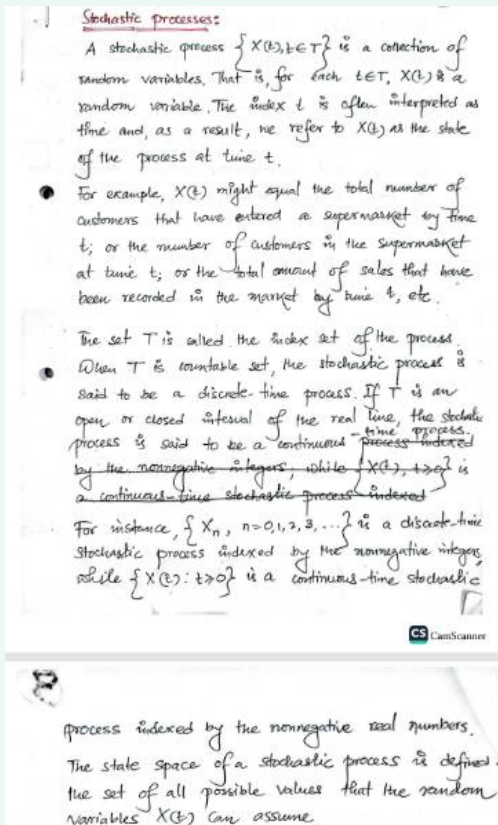


Explain stochastic processes



explain this a very very clearly

This image explains **stochastic processes**, so let me break it down for you step by step:

1. Stochastic Process:

- A **stochastic process** is a collection of **random variables** indexed by time. This means for each time t , you have a random variable $X(t)$. These random variables change over time, representing the **state** of the process at each point in time.
- **For example:**
 - $X(t)$ could represent the number of customers that entered a store by time t ,
 - The total sales made by a store by time t ,

- Or the number of items sold by time t , etc.
- In short, a stochastic process is a way to model a system that evolves over time in a random way.

2. The Index Set (T):

- The set T is called the **index set** and represents the **time** over which the process is observed.
- If T is a **countable set** (like $\{1, 2, 3, \dots\}$), the process is called a **discrete-time process**.
- If T is an **open or closed interval of real numbers** (like $[0, \infty)$, $(-\infty, \infty)$), the process is called a **continuous-time process**.
- **Example:**
 - A **discrete-time stochastic process** would look like $\{X_n, n = 0, 1, 2, 3, \dots\}$, which is indexed by the integers (discrete values).
 - A **continuous-time stochastic process** would look like $X(t)$, where t can take any real value, not just integers. This can represent processes that evolve smoothly over time, such as stock prices.

3. State Space of a Stochastic Process:

- The **state space** of a stochastic process is the set of all possible values that the random variables $X(t)$ can take.
 - For example, if $X(t)$ represents the number of customers in a store, the state space would include all possible counts of customers (like 0, 1, 2, etc.).
 - If $X(t)$ represents sales amounts, the state space would include all possible sales values (such as 0, 1.5, 1000, etc.).

In short, this concept helps us understand systems that evolve randomly over time, such as stock prices, population growth, or the number of people entering a shop. The two main types of stochastic processes are **discrete-time** and **continuous-time** processes, depending on whether time is treated as a discrete set of points or a continuous flow.



Stochastic processes

A stochastic process $\{X(t), t \in T\}$ is a collection of random variables. That is, for each $t \in T$, $X(t)$ is a random variable. The index t is often interpreted as time and, as a result, we refer to $X(t)$ as the state of the process at time t .

- For example, $X(t)$ might equal the total number of customers that have entered a supermarket by time t ; or the number of customers in the supermarket at time t ; or the total amount of sales that have been recorded in the market by time t , etc.

The set T is called the index set of the process.

- When T is countable set, the stochastic process is said to be a discrete-time process. If T is an open or closed interval of the real line, the stochastic process is said to be a continuous-time process indexed by the nonnegative integers, while $\{X(t), t \geq 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

For instance, $\{X_n, n = 0, 1, 2, 3, \dots\}$ is a discrete-time stochastic process indexed by the nonnegative integers, while $\{X(t), t \geq 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

CS CamScanner

process indexed by the nonnegative real numbers.

The state space of a stochastic process is defined as the set of all possible values that the random variables $X(t)$ can assume.

explain very clearly

This image explains **stochastic processes**, and I'll explain it step by step with clear examples.

1. What is a Stochastic Process?

- A **stochastic process** is a collection of **random variables** indexed by time.
- For each time t , $X(t)$ is a random variable.
- Index t** is typically interpreted as time. So, $X(t)$ represents the state of the process at that particular time.

Example:

- $X(t)$ could be the total number of customers entering a store by time t , the number of customers in the store at time t , or the total amount of sales made by time t .
- The process evolves over time, and $X(t)$ changes randomly over time.

2. The Index Set T :

- The set T is called the **index set** and represents the time at which you observe the process.
- If T is **countable** (e.g., $\{1, 2, 3, \dots\}$), the stochastic process is called a **discrete-time process**.
- If T is a **continuous interval** of real numbers (e.g., $[0, \infty)$, or any real interval), the process is called a **continuous-time process**.

Example:

- A **discrete-time process** might involve observing a system at specific time steps, like every minute, hour, or day (e.g., $X_1, X_2, X_3, \dots$).
 - A **continuous-time process** can take any real value for t , meaning it can be observed at any time, even between the discrete time points. For instance, the price of a stock at any given second.
-

3. Discrete vs. Continuous Stochastic Processes:

- A **discrete-time process** could be written as $\{X_n, n = 0, 1, 2, 3, \dots\}$, where n is a countable index (e.g., the day number).
- A **continuous-time process** could be written as $X(t), t \geq 0$, where t can take any real number (e.g., $t = 1.5, t = 2.7$, etc.).

Example:

- $\{X_n, n = 0, 1, 2, 3, \dots\}$ is a discrete-time process because it only takes values at specific times (e.g., at the end of each day).
 - $X(t), t \geq 0$ is a continuous-time process because it can take values at any time t within the range of possible times (e.g., you can track a stock price at any point in time).
-

4. State Space of a Stochastic Process:

- The **state space** refers to the set of all possible values that the random variables $X(t)$ can take.

- For example:
 - If $X(t)$ represents the number of people entering a store, the state space would be all possible numbers of customers.
 - If $X(t)$ represents sales amounts, the state space would include all possible values of sales (from 0 to any positive number).

Summary:

- A **stochastic process** is a model for a system that evolves randomly over time.
- **Discrete-time processes** are observed at specific time steps, whereas **continuous-time processes** are observed at any point in time.
- The **state space** is the range of possible outcomes for the process.

This framework is used to model various systems like stock prices, population growth, queueing systems, and many other real-world phenomena that evolve randomly.



explain

This image explains **parametric families of univariate probability distributions**. Let me break it down for you clearly:

1. Parametric Family of Distributions:

- A **parametric family of distributions** is a set of probability distributions that are indexed by **parameters**.
- These **parameters** control the shape or characteristics of the distribution.
- Each distribution in the family has a **standard name** and can be derived based on specific assumptions about the random variables involved.

Example:

- The **Normal distribution** is part of a parametric family where the parameters are the **mean (μ)** and **variance (σ^2)**.

2. Mathematical Treatment:

- These distributions can be represented by **algebraic functions**, which allows them to be analyzed and treated mathematically.
- This means you can apply **statistical methods** to study them, calculate probabilities, and perform other mathematical operations.

3. Discrete vs. Continuous Distributions:

- The parametric distributions discussed here include both **discrete** and **continuous** types of distributions.

4. Discrete Distributions:

- **Discrete distributions** are used to model situations where the random variable takes discrete values (like counting numbers).

The most common discrete distributions include:

- **Bernoulli distribution:** Models the outcome of a single trial with two possible outcomes (success or failure).
- **Binomial distribution:** Models the number of successes in a fixed number of independent Bernoulli trials.

- **Hypergeometric distribution:** Models the number of successes in a sample drawn without replacement from a finite population.
 - **Poisson distribution:** Models the number of events occurring in a fixed interval of time or space, with the events occurring independently and at a constant rate.
 - **Geometric distribution:** Models the number of trials needed to get the first success in a series of independent Bernoulli trials.
 - **Negative binomial distribution:** Generalizes the geometric distribution to model the number of trials needed to achieve a fixed number of successes.
 - **Multinomial distribution:** Generalizes the binomial distribution for experiments with more than two possible outcomes.
-

5. Continuous Distributions:

- **Continuous distributions** are used to model situations where the random variable can take any value within a certain range (such as real numbers).

The most commonly discussed continuous distributions are:

- **Normal distribution:** Describes a bell-shaped curve, which is widely used to model data that clusters around a mean.
 - **Exponential distribution:** Models the time between events in a Poisson process, often used to model waiting times.
 - **Beta distribution:** Used to model probabilities that are constrained between 0 and 1, often used in Bayesian statistics.
 - **Gamma distribution:** Generalizes the exponential distribution and is often used in queuing models.
 - **Rectangular distribution:** Also known as the **uniform distribution**, it models situations where all outcomes are equally likely within a specific range.
-

Summary:

- **Parametric families** of distributions consist of a set of distributions, each controlled by one or more parameters.
- These distributions can be **discrete** or **continuous** and are important for modeling and decision-making in statistics and probability.

- Understanding these distributions helps in analyzing real-world random processes, such as counting events, measuring waiting times, and modeling various phenomena like heights, test scores, or financial data.



Bernoulli Distribution : / Bernoulli Process :

A considerable group of random phenomena, known as Bernoulli process, named after James Bernoulli (1654-1705), follows the simplest probability distribution - involving only two possible outcomes, such as head or tail, success or failure, defective or non-defective, smoker or non-smoker, married or unmarried and like that.

To standardize the terminology describing these and many other similar processes, we would call one of the possible outcomes a "success" and the other a "failure". These names are used only to identify the outcomes and bear no connotation of success or failure in real life.

Customarily, the outcome of primary interest in a study is labeled a success (even if it is a disastrous event).

In a study of prevalence of marriage dissolution or rate of electricity failure, the marital status "divorced" or "load shedding" may be attributed the statistical name "success".

explain very clearly

This image explains the **Bernoulli Distribution/Process**. Let me explain it clearly in steps.

1. Bernoulli Process:

- The **Bernoulli process** (named after James Bernoulli) is the simplest **probability distribution** involving **two possible outcomes**.
- These outcomes can be something like:
 - **Head or Tail** (in a coin flip),
 - **Success or Failure** (e.g., passing or failing a test),
 - **Defective or Non-defective** (in manufacturing),
 - **Married or Unmarried** (in a survey), etc.

2. Standardization of Terminology:

- To standardize how we talk about the outcomes of processes, **one outcome** is called a "**success**" and the other a "**failure**".
- These terms don't necessarily mean positive or negative—they are just labels to distinguish the two possible outcomes.
- For example:
 - If we are studying a survey on **marriage dissolution**, the outcome "**divorced**" might be considered a "success" in this case (because it's the event of interest).
 - Similarly, "**failure**" is just a label to describe the other outcome, such as "non-divorced".

3. Bernoulli Trials:

- The **Bernoulli trials** are the **independent repetitions** of a random experiment that only have two possible outcomes: success or failure.
- **For example:**
 - **Coin Toss:** You toss a coin repeatedly. If it lands on **heads**, it could be labeled as "success", and if it lands on **tails**, it could be labeled as "failure".
 - **Manufacturing:** In a factory, if an item produced is **defective**, it might be labeled as "failure", and if it is **non-defective**, it might be labeled as "success".

4. Bernoulli Trial Examples:

i. Coin Toss Example:

- Suppose you toss a fair coin repeatedly. If it shows heads, you can label that as **success** (1), and if it shows tails, you can label it as **failure** (0).
- The random variables for these tosses would be $X_1, X_2, X_3, \dots$, and the outcome of each toss can be either 0 or 1 (heads or tails).
- Since the coin is fair, the **probability** of heads (success) is 0.5, and the probability of tails (failure) is also 0.5.
- These trials follow a **Bernoulli process** with parameter $p = 0.5$.

ii. Manufacturing Example:

- In a factory, 20% of the items produced are defective. You inspect each item randomly and label it as:
 - **Success (0)** if the item is **non-defective**,
 - **Failure (1)** if the item is **defective**.
- This results in a sequence of trials $X_1, X_2, X_3, \dots$ where each trial corresponds to checking an item, and the outcome is either "success" or "failure".
- The probability of success (non-defective) is 0.8, and the probability of failure (defective) is 0.2.
- These trials form a **Bernoulli process** with parameter $p = 0.2$.

Key Points:

- A **Bernoulli process** involves repeating an experiment with only two possible outcomes (success or failure).
- Each trial is independent, and the outcomes are random.
- The term **Bernoulli trial** refers to each individual repetition of the experiment.
- The **probability of success** and **probability of failure** are constant across trials.



Defⁿ of Bernoulli distribution:

A random variable X is said to have a Bernoulli distribution with parameter p ($0 \leq p \leq 1$) if X can take on only the values 0 and 1 and the probabilities are

$$\left. \begin{aligned} P(X=1) &= p \\ \& \ P(X=0) &= 1-p \end{aligned} \right\} \text{--- ①}$$

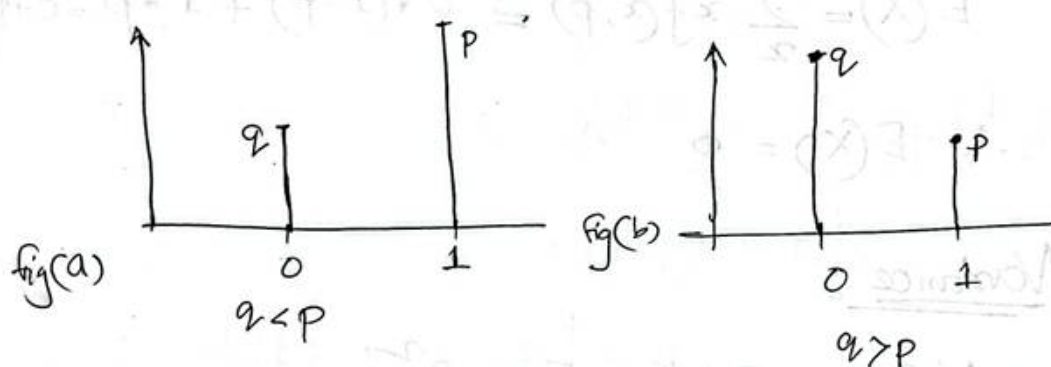
Since the distribution involves only two cases of events, it is also known as two-point probability distribution.

If we let $q = 1-p$, then the probability function of X can be written as follows:

$$f(x; p) = \begin{cases} p^x q^{1-x} & ; \text{ for } x = 0, 1 \\ 0 & ; \text{ otherwise} \end{cases}$$

To verify that the probability function (2) actually represents the Bernoulli distribution specified by the probabilities (1), it is necessary to note that $f(1, p) = p$ and $f(0, p) = q$.

The tabular representation of the random variable X is as follows:



Bernoulli functions with $q < p$ (Fig-a) & $q > p$ (Fig-b)

x	$P(X=x)$
0	$q = 1 - p$
1	p

explain very clearly

These images explain the **Bernoulli Distribution** and its properties. Here's a breakdown of the key points:

1. Definition of Bernoulli Distribution:

- A **random variable** X is said to have a **Bernoulli distribution** with parameter p (where $0 \leq p \leq 1$) if X can take only two possible values: **0** or **1**.
 - $P(X = 1) = p$ (the probability of success),
 - $P(X = 0) = 1 - p$ (the probability of failure).

This is often called a **two-point probability distribution** because there are only two outcomes: success (1) and failure (0).

2. Probability Function:

The probability function for a Bernoulli distribution is written as:

$$f(x; p) = \begin{cases} p^x q^{1-x} & \text{for } x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

where $q = 1 - p$, and p is the probability of success (1), and q is the probability of failure (0).

- For $x = 1$, $f(x) = p$,
 - For $x = 0$, $f(x) = q$.
-

3. Verification of the Probability Function:

- To verify that the given probability function represents the Bernoulli distribution, you check that:
 - $f(1; p) = p$,
 - $f(0; p) = q$, where $q = 1 - p$.
-

4. Graphical Representation:

- The **graphical representation** of the Bernoulli distribution is shown in two figures:
 - **Fig. (a):** When $q < p$, the height at $x = 1$ (the success point) is **higher** than at $x = 0$ (the failure point).

- **Fig. (b):** When $q > p$, the height at $x = 0$ (failure) is **higher** than at $x = 1$ (success).

These graphs represent the probability distribution where:

- The height of the bar at **0** corresponds to q ,
- The height of the bar at **1** corresponds to p .

5. Tabular Representation:

The table below shows the Bernoulli distribution for the random variable X :

- If $X = 0$, then the probability $P(X = 0) = q$.
- If $X = 1$, then the probability $P(X = 1) = p$.

Where:

- $q = 1 - p$.

Summary:

- The **Bernoulli distribution** is a **binary distribution** where the random variable can take only two values: 0 (failure) or 1 (success).
- The probability of success is p , and the probability of failure is $1 - p$.
- The probability function is defined and can be represented as $f(x; p)$.
- **Graphical and tabular representations** are used to understand the distribution visually and numerically.



Properties of Bernoulli Distribution:

Similar to other distributions, the Bernoulli distribution has its mean, variance, and other descriptive measures. These are obtained as follows:

Mean:

$$E(X) = \sum_{x=0}^1 x f(x, p) = 0 \cdot (1-p) + 1 \cdot p = 0 + p = p$$

$\therefore E(X) = p$

Variance:

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \sum_{x=0}^1 x^2 f(x, p) - [E(X)]^2$$

$$= 0^2 \cdot (1-p) + 1^2 \cdot p - (p)^2$$

$$= p - p^2$$

$$= p(1-p)$$

$$= pq \quad [\because 1-p=q]$$

$\therefore V(X) = pq$

explain

This image explains the **properties of the Bernoulli distribution**, specifically focusing on the **mean** and **variance**. Let's go through the explanation step by step.

1. Mean of Bernoulli Distribution:

- The **mean** (or expected value) of a random variable X with a Bernoulli distribution is calculated as:

$$E(X) = \sum_x x f(x, p)$$

Where:

- $f(x, p)$ is the probability mass function of the Bernoulli distribution.
- For a Bernoulli distribution, the random variable X can take two values: **0** (failure) or **1** (success).
- The probability of success is p , and the probability of failure is $1 - p$.

The calculation is as follows:

$$E(X) = 0 \cdot (1 - p) + 1 \cdot p = p$$

So, the **mean** (expected value) of the Bernoulli distribution is:

$$E(X) = p$$

2. Variance of Bernoulli Distribution:

- The **variance** of a random variable X is calculated using the formula:

$$V(X) = E(X^2) - [E(X)]^2$$

Where:

- $E(X^2)$ is the expected value of X^2 ,
- $E(X)$ is the mean of X .

The calculation for variance proceeds as follows:

- First, we calculate $E(X^2)$:

$$E(X^2) = \sum_x x^2 f(x, p)$$

Since X only takes the values 0 and 1, we have:

$$E(X^2) = 0^2 \cdot (1 - p) + 1^2 \cdot p = p$$

- Now, we calculate $V(X)$:

$$V(X) = E(X^2) - [E(X)]^2 = p - p^2$$

We can factor this as:

$$V(X) = p(1 - p)$$

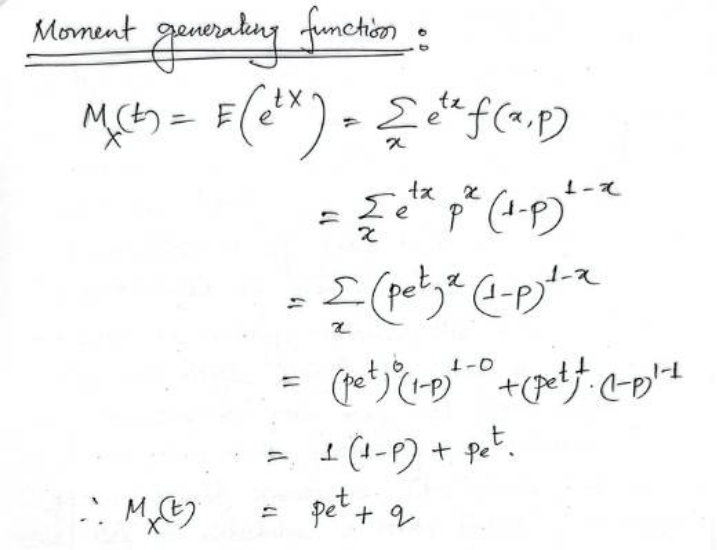
So, the **variance** of the Bernoulli distribution is:

$$V(X) = p(1 - p)$$

Summary:

- The **mean** (expected value) of a Bernoulli distribution is $E(X) = p$.
- The **variance** of a Bernoulli distribution is $V(X) = p(1 - p)$.





Handwritten derivation of the Moment Generating Function (MGF) for the Bernoulli distribution:

$$\begin{aligned}
 \text{Moment generating function:} \\
 M_X(t) &= E(e^{tX}) = \sum_x e^{tx} f(x, p) \\
 &= \sum_x e^{tx} p^x (1-p)^{1-x} \\
 &= \sum_x (pe^t)^x (1-p)^{1-x} \\
 &= (pe^t)^0 (1-p)^{1-0} + (pe^t)^1 (1-p)^{1-1} \\
 &= 1(1-p) + pe^t \\
 \therefore M_X(t) &= pe^t + q
 \end{aligned}$$

This image explains the **Moment Generating Function (MGF)** for the **Bernoulli Distribution**. Let's break it down clearly:

Moment Generating Function (MGF):

The **Moment Generating Function (MGF)** of a random variable X is defined as:

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x, p)$$

Where:

- E denotes the expected value,
- e^{tx} is the exponential function raised to the power of tx ,
- $f(x, p)$ is the probability mass function of the Bernoulli distribution.

Step-by-Step Derivation:

1. Substitute the Probability Mass Function:

The probability mass function for the Bernoulli distribution is:

- $P(X = 1) = p$
- $P(X = 0) = 1 - p$

The MGF formula becomes:

$$M_X(t) = \sum_x e^{tx} \cdot p \cdot (1-p)^{1-x}$$

2. Expand the Sum:

Since x can be either 0 or 1, we expand the sum for both cases of x :

$$M_X(t) = e^{t \cdot 0} \cdot p \cdot (1-p) + e^{t \cdot 1} \cdot (1-p) \cdot p$$

3. Simplify the Expression:

- For $x = 0$, $e^{t \cdot 0} = 1$, so the first term becomes $1 \cdot (1-p)$.
- For $x = 1$, $e^{t \cdot 1} = e^t$, so the second term becomes $e^t \cdot p$.

The result is:

$$M_X(t) = (1-p) + pe^t$$

4. Final Form of the MGF:

Therefore, the Moment Generating Function for the Bernoulli distribution is:

$$M_X(t) = pe^t + (1-p)$$

Summary:

- The **Moment Generating Function (MGF)** for a **Bernoulli distribution** is $M_X(t) = pe^t + (1-p)$.
- The MGF is used to generate the moments (like mean and variance) and can also be useful in analyzing the properties of the distribution.



Chebyshev's Theorem / Chebyshev's Inequality:

If the probability or population histogram is roughly bell-shaped and the mean and variance are known, the empirical rule (i.e., $\mu \pm \sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$ contain respectively almost 68%, 95% and 99.99% of the measurements) is of great help in approximating the probabilities of certain intervals.

• However in many instances, the shapes of probability histograms differ significantly from a mound shape, and the empirical rule may not yield useful approximations to the probability of interest. The following result, known as Chebyshev's theorem, can be used to determine a lower bound for the probability that the random variable X of interest falls in an interval $\mu \pm k\sigma$.

explain

This image explains **Chebyshev's Theorem** (also known as **Chebyshev's Inequality**). Here's a clear breakdown:

1. When the Empirical Rule is Not Sufficient:

- The **empirical rule** (or **68-95-99.7 rule**) applies when the probability distribution is **bell-shaped** (like a normal distribution) and the **mean** (μ) and **variance** (σ^2) are known.
- According to the empirical rule:
 - **68%** of the data falls within $\mu \pm 1\sigma$,
 - **95%** falls within $\mu \pm 2\sigma$,
 - **99.7%** falls within $\mu \pm 3\sigma$.
- This is a quick way to estimate the probability that a random variable falls within certain intervals for **normally distributed data**.

2. Limitations of the Empirical Rule:

- However, not all distributions follow a **bell-shaped curve**. In such cases, the **empirical rule** may not be helpful.
- When the distribution is **not bell-shaped** (i.e., it's skewed or has heavy tails), the empirical rule might not give accurate probabilities.

3. Chebyshev's Theorem:

- **Chebyshev's Theorem** (or **Chebyshev's Inequality**) is used in cases where the distribution is not necessarily normal or bell-shaped.
- It provides a **lower bound** on the probability that a random variable X falls within an interval around the **mean** μ (with some distance defined in terms of standard deviation σ).
- **Chebyshev's Inequality** states that for any distribution, regardless of its shape:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

This inequality says that the proportion of values that are **at least k standard deviations away** from the mean is at most $\frac{1}{k^2}$.

4. Interpreting Chebyshev's Inequality:

- For **any** $k > 1$, the probability that a value lies within k standard deviations of the mean is at least:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

- For example:
 - For $k = 2$, at least **75%** of the data falls within $\mu \pm 2\sigma$,
 - For $k = 3$, at least **89%** of the data falls within $\mu \pm 3\sigma$.

5. Key Takeaways:

- **Chebyshev's Theorem** is a more general inequality that works for all distributions, **not just normal distributions**.
- It provides a **guaranteed lower bound** on the proportion of data within a certain range from the mean, regardless of the data's shape.



what is standard deviation

Standard deviation is a statistical measure that quantifies the amount of variability or spread in a set of data points. In simpler terms, it shows how much the individual data points deviate from the mean (average) of the dataset.

Here's a clear explanation:

How to Interpret Standard Deviation:

- A **low standard deviation** means the data points are **close** to the mean, indicating **less variation**.
 - A **high standard deviation** means the data points are **spread out** over a larger range, indicating **more variation**.
-

Formula for Standard Deviation:

For a population, the standard deviation σ is calculated using the formula:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Where:

- σ = standard deviation,
- N = number of data points,
- x_i = each data point,
- μ = mean of the data,
- $\sum$ = sum of the values from 1 to N .

For a **sample** (subset of a population), the formula is slightly different to account for the smaller data size:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Where:

- s = sample standard deviation,

- n = sample size,
 - $\bar{x}$ = sample mean.
-

Steps to Calculate Standard Deviation:

1. Find the Mean:

- Add all the data points and divide by the number of data points.

2. Subtract the Mean:

- For each data point, subtract the mean and square the result. This gives the squared deviations.

3. Average the Squared Deviations:

- Sum up the squared deviations and divide by the number of data points (or $n - 1$ for a sample).

4. Take the Square Root:

- The square root of this average gives you the standard deviation.
-

Example:

Consider the data points: 2, 4, 4, 4, 5, 5, 7, 9.

1. Calculate the Mean:

$$\mu = \frac{2 + 4 + 4 + 4 + 5 + 5 + 7 + 9}{8} = 5$$

2. Subtract the Mean and Square the Result:

- $(2 - 5)^2 = 9$
- $(4 - 5)^2 = 1$
- $(4 - 5)^2 = 1$
- $(4 - 5)^2 = 1$
- $(5 - 5)^2 = 0$
- $(5 - 5)^2 = 0$
- $(7 - 5)^2 = 4$

- $(9 - 5)^2 = 16$

3. Sum the Squared Deviations:

$$9 + 1 + 1 + 1 + 0 + 0 + 4 + 16 = 32$$

4. Calculate the Average (Variance):

$$\text{Variance} = \frac{32}{8} = 4$$

5. Take the Square Root of the Variance:

$$\sigma = \sqrt{4} = 2$$

So, the **standard deviation** is 2.

Conclusion:

Standard deviation is a key concept in statistics because it gives a measure of how spread out or clustered the data is around the mean. A larger standard deviation means more spread, while a smaller standard deviation indicates that the data is more concentrated around the mean.